

1. Bellman Optimality Equation
3. State-Action-Reward-State-Action
5. Semantic Segmentation
7. Bias<sup>2</sup> + Variance + Irreducible Error

$$9. \left\lfloor \frac{128-10}{2} + 1 \right\rfloor = \lfloor 59 + 1 \rfloor = 60$$

Output shape: (60, 60, 5)

11. Density-Based Spatial Clustering of Applications with Noise

13. 1 (because of softmax normalization)

15. Backpropagation Through Time

17. Expected Cumulative Reward

19.  $7 \times 7$

True/False answers

- |         |          |           |         |          |         |          |
|---------|----------|-----------|---------|----------|---------|----------|
| 1. True | 2. False | 3. True   | 4. True | 5. False | 6. True | 7. False |
| 8. True | 9. True  | 10. False |         |          |         |          |

Long-form answers

1.

$$-1 \times 20 + 3 \times 5 + (-2) \cdot 30 + 2 \cdot 0$$

$$-1 \cdot 70 + 3 \cdot 200 + (-2) \cdot 0 + 2 \cdot 100$$

$$-1 \times 70 + 3 \times 60 + (-2) \cdot 170 + 2 \cdot 160$$

$$-1 \cdot 210 + 3 \cdot 0 + (-2) \cdot 5 + 2 \cdot 1$$

$$-1 \times 90 + 3 \cdot 0 + (-2) \cdot 1 + 2 \cdot 20$$

$$-1 \cdot 10 + 3 \cdot 100 + (-2) \cdot 80 + 2 \cdot 5$$

$$-65 \quad 730$$

$$-230 \quad -218$$

$$-52 \quad 140$$

Applying ReLU, we get

$$0 \quad 730$$

$$90 \quad 0$$

$$0 \quad 140$$

Applying average pooling, we get

$$(0 + 730) / 2 = 365$$

$$(90 + 0) / 2 = 45$$

$$(0 + 140) / 2 = 70$$

2. Forward pass:

$$\text{net } h_1 = 0.4 \cdot 2 + (-0.1) \cdot (-1) = 0.9$$

$$\text{net } h_2 = 0.6 \cdot 2 + (-0.3) \cdot (-1) = 1.5$$

$$\text{sigmoid}(\text{net } h_1) = \frac{1}{1 + e^{-0.9}} \approx 0.7109$$

$$\text{sigmoid}(\text{net } h_2) = \frac{1}{1 + e^{-1.5}} \approx 0.8176$$

2. Huber Loss

4. Forget Gate

6. Leaky ReLU

8. Each filter has  $10 \times 10 \times 3 = 300$  weights  
~~weights~~ + 1 bias = 301, for 5 filters  
 $5 \times 301 = 1505$  parameters

10. 2 gates (Reset gate and Update gate)

12. Faster R-CNN

14. Policy Gradient

16. Non-maximum Suppression

18. Slack Variable

20. Weight Decay

91

-6

$$\text{net}_{o1} = 0.5 \cdot 0.1109 + (-0.2) \cdot 0.8176 = 0.19193$$

$$\text{sigmoid}(\text{net}_{o1}) = \frac{1}{1 + e^{-0.19193}} \approx 0.5478$$

Backward pass:

$$\delta^{o1} = (0.5478 - 1) \cdot 0.5478 \cdot (1 - 0.5478) \approx -0.1120$$

$$\delta^{h1} = 0.7109 \cdot (1 - 0.7109) \cdot 0.5 \cdot (-0.1120) \approx -0.0115$$

$$\delta^{h2} = 0.8176 \cdot (1 - 0.8176) \cdot (-0.2) \cdot (-0.1120) \approx 0.0033$$

Updating weights

$$w_{h1o1} = 0.5 - (-0.1120 \cdot 0.7109) \approx 0.5796$$

$$w_{h2o1} = -0.2 - (-0.1120 \cdot 0.8176) \approx -0.1084$$

$$w_{x1h1} = 0.4 - (-0.0115 \cdot 2) \approx 0.377$$

$$w_{x1h2} = 0.6 - (0.0033 \cdot 2) = 0.5934$$

$$w_{x2h1} = -0.1 - (-0.0115 \cdot -1) = -0.0885$$

$$w_{x2h2} = -0.3 - (0.0033 \cdot -1) = -0.2967$$

$$3. f^t = \text{sigmoid}(-2.5 \cdot 0.1 + (-0.8) \cdot 2.3) + 2.1 \approx 0.5028$$

$$i^t = \text{sigmoid}(-2.5 \cdot (-1.4) + (-0.8) \cdot (-1.1) - 1.7) \approx 0.9359$$

$$c^t = \tanh(-2.5 \cdot 0.9 + 0.5 \cdot (-0.5) + 1.8) \approx -0.0499$$

$$o^t = \text{sigmoid}(-2.5 \cdot (-0.8) + 1.6 - 2.6) \approx 0.1323$$

$$c^t = f^t \cdot c^{t-1} + i^t \cdot c^t$$

$$= 0.5028 \cdot 1.7 + 0.9359 \cdot (-0.0499) \approx 0.8562$$

$$h^t = o^t \cdot \tanh(c^t) = 0.1323 \cdot \tanh(0.8562) \approx 0.0669$$

$$4. P(\text{Positive}) = \frac{7}{8} \quad P(\text{Negative}) = \frac{3}{8} \quad P(\text{Neutral}) = \frac{2}{8}$$

~~$$P(w|\text{Positive}) = \frac{\text{count}(w)}{\text{total words}}$$~~

Positive Total words: 15

Negative Total words: 13

Neutral Total words: 16

Words: the, movie, theater, is, costly

Vocabulary size  $|V| = 17$

$$P(w|\text{Positive}) = \frac{\text{count}(w) + 1}{\text{total words} + |V|} = \frac{\text{count} + 1}{15 + 17} = \frac{\text{count}(w) + 1}{32}$$

$$\text{the: } \frac{0+1}{32}$$

$$\text{movie: } \frac{2+1}{32}$$

$$\text{theater: } \frac{2+1}{32}$$

$$\text{is: } \frac{3+1}{32}$$

$$\text{costly: } \frac{0+1}{32}$$

$$P(w|\text{Negative}) = \frac{\text{count}(w) + 1}{13 + 17} = \frac{\text{count}(w) + 1}{30}$$

$$\text{the: } \frac{2+1}{30}$$

$$\text{movie: } \frac{2+1}{30}$$

$$\text{theater: } \frac{2+1}{30}$$

$$\text{is: } \frac{3+1}{30}$$

$$\text{costly: } \frac{1+1}{30}$$

$$P(D|\text{Positive}) = \frac{1}{32} \cdot \frac{3}{32} \cdot \frac{3}{32} \cdot \frac{4}{32} \cdot \frac{1}{32} = \frac{36}{325} \approx 4.02 \times 10^{-8}$$

$$P(D|\text{Negative}) = \frac{4 \cdot 3 \cdot 3 \cdot 4 \cdot 2}{30^5} = \frac{288}{30^5} \approx 2.05 \times 10^{-8}$$

$$p(w|Neutral) = \frac{\text{count}(w)+1}{16+17} = \frac{\text{count}(w)+1}{33}$$

$$\begin{aligned} \text{the: } & \frac{2+1}{33} \\ \text{motion: } & \frac{2+1}{33} \\ \text{therefor: } & \frac{2+1}{33} \\ \text{is: } & \frac{2+1}{33} \\ \text{costly: } & \frac{1+1}{33} \end{aligned}$$

$$p(D|Neutral) = \frac{3 \cdot 3 \cdot 3 \cdot 3 \cdot 2}{33^5} = \frac{162}{33^5} \approx 1.07 \times 10^{-6}$$

~~Major~~ Label: Negative.

$$S. P(\text{Yes}) = \frac{9}{16} \quad P(\text{No}) = \frac{7}{16}$$

~~P(Dog|Yes)~~ ~~For Yes:~~ ~~Animal=Dog~~ ~~Size=Small~~ ~~Body Color=White~~

Size of Animal: Likelihoods (with  $\alpha=1$ , total categorize per feature = 3)

| Feature     | Value | Count in Yes | Count in No |
|-------------|-------|--------------|-------------|
| Animal=Dog  | 4     | 1            | 0           |
| Size=Small  | 1     | 2            | 0           |
| Color=White | 1     | 3            | 0           |

$$P(\text{Dog}|\text{Yes}) = \frac{4+1}{9+3} = \frac{5}{12} \quad P(\text{Small}|\text{Yes}) = \frac{2+1}{9+3} = \frac{3}{12} \quad P(\text{White}|\text{Yes}) = \frac{1+1}{9+3} = \frac{2}{12}$$

$$P(\text{Yes}|\text{Feature}) \propto \frac{9}{16} \cdot \frac{5}{12} \cdot \frac{3}{12} \cdot \frac{2}{12} = \frac{9 \cdot 5 \cdot 4 \cdot 2}{16 \cdot 12^3} \approx 0.013$$

For No

$$P(\text{Dog}|\text{No}) = \frac{1+1}{7+3} = \frac{2}{10} \quad P(\text{Small}|\text{No}) = \frac{2+1}{7+3} = \frac{3}{10} \quad P(\text{White}|\text{No}) = \frac{3+1}{7+3} = \frac{4}{10}$$

$$P(\text{No}|\text{Feature}) \propto \frac{7}{16} \cdot \frac{2}{10} \cdot \frac{3}{10} \cdot \frac{4}{10} = \frac{7 \cdot 2 \cdot 3 \cdot 4}{16 \cdot 1000} \approx 0.0105$$

(a): Prediction: Yes

(b): Feature Value Count in Yes

|             |   |   |
|-------------|---|---|
| Animal=Cow  | 3 | 2 |
| Size=Medium | 2 | 1 |
| Color=Brown | 1 | 3 |

$$P(\text{Cow}|\text{Yes}) = \frac{3+1}{9+3} = \frac{4}{12} \quad P(\text{Medium}|\text{Yes}) = \frac{2+1}{9+3} = \frac{3}{12} \quad P(\text{Brown}|\text{Yes}) = \frac{1+1}{9+3} = \frac{2}{12}$$

$$P(\text{Yes}|\text{Feature}) \propto \frac{9}{16} \cdot \frac{4}{12} \cdot \frac{3}{12} \cdot \frac{2}{12} \approx 0.0078$$

$$P(\text{Cow}|\text{No}) = \frac{2+1}{7+3} = \frac{3}{10} \quad P(\text{Medium}|\text{No}) = \frac{1+1}{7+3} = \frac{2}{10} \quad P(\text{Brown}|\text{No}) = \frac{3+1}{7+3} = \frac{4}{10}$$

$$P(\text{No}|\text{Feature}) \propto \frac{7}{16} \cdot \frac{3}{10} \cdot \frac{2}{10} \cdot \frac{4}{10} \approx 0.0105$$

Prediction: No